

Ryan Peach

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Objective: Senior specialist in AI and Cloud Technologies seeking meaningful, impact-driven work.

EDUCATION

Georgia Tech	Atlanta, GA
<i>MS in Computer Science, Machine Learning Specialty. Computer Vision & Robotics Equiv. Coursework</i>	
<i>2015 – 2018</i>	
Western Kentucky University	Bowling Green, KY
<i>BS in Electrical Engineering, Minor in Mathematics</i>	
<i>2010 – 2014</i>	

MAJOR SKILLS

<i>Python Scientific Stack, Data Science</i>	<i>Expert, 2013-Present</i>
<i>Docker, Kubernetes, AWS</i>	<i>Expert, 2017-Present</i>
<i>Machine Learning: Deep Learning, Reinforcement Learning</i>	<i>MS, 2015-Present</i>
<i>Large Language Models: Safety, Architecture, API Tools, Open Source Development</i>	<i>Advanced, 2022-Present</i>
<i>Engineering: 3D Printing, PCB Design, OnShape</i>	<i>BS, 2010-Present</i>
<i>Rust Language</i>	<i>Advanced, 2019-Present</i>

EXPERIENCE

Senior Principal Engineer	2023-Present
<i>Ansys / Synopsys</i>	<i>Atlanta, GA</i>
<i>Architecture</i>	

- Trained management on transition from single team Kubernetes infrastructure to enterprise infrastructure.
- Architected globally available infrastructure, designed and maintained authoritative infrastructure diagrams covering approximately dozens of interoperating services.
- Designed GraphQL, gRPC, and OpenAPI APIs for core backend services.

Large Language Models

- Participated in early open source efforts with LangChain, LlamaIndex, etc. Contributing code to repos.
- Advised on company LLM projects across multiple teams, attended conferences on LLM research

Director of Cloud DevOps	2021-2023
<i>OnScale</i>	<i>Atlanta, GA</i>
<i>Architecture</i>	

- Managed custom Kubernetes controller which managed simulation jobs on high-performance compute (HPC) infrastructure.
- Refactored and modernized Infrastructure as Code (IaC).
- Managed AWS pricing, SEV0 outages, observability, and data analytics

Machine Learning

- Managed team and cloud resources for development of ML Pipeline which predicted simulation time based on simulation metrics prior to runtime, giving customer an estimate of pricing.
- ML Pipeline needed to automatically retrain on new data periodically so architected ETL and SageMaker training

Management

- Completely up and running within 2 weeks of hiring due to leadership transition.
- Directed all employees and decisions on Backend, Architecture, and ML projects.
- Managed ML, HPC, Data, and DevOps teams during Ansys acquisition.
- Transitioned to technical track after transitioning all teams to new management under Ansys.

Machine Learning Researcher - Lead Architect

Keysight Technologies

2017 – 2021

Atlanta, GA

Architecture

- Built a CloudFormation stack using ECR, Lambda that spins up thousands of Docker containers to train, tune, and select optimal SVG models. Weeks of compute completed in an hour regardless of scale.
- Leading the team in code reviews, managed all the DevOps for the team including managing the git repo and the Jenkins CI/CD.
- Designed PC cluster hardware and managed on-prem Unix, Spark, PostgreSQL, MongoDB, and Kubernetes server.

Machine Learning

- Independently investigated almost a dozen potential ML applications at the company, providing preliminary results and feasibility studies.
- Wrote approximately 7 internal publications on patentable inventions, as well as giving multiple company-wide presentations on my work.
- Implemented large scale hyperparameter tuning systems on AWS. Some custom and others using SageMaker. Worked with AutoML solutions.
- Applied Reinforcement Learning to multiple novel problem domains using Monte Carlo Tree Search and Q Learning.
- Data visualization and web-enabled dashboard creation in Tensorboard, Plotly, Dash, Django, Tableau (similar software), Matplotlib, Altair, and Jupyter.

Leadership

- Promoted from Developer to Researcher within only 1 year due to outstanding performance. Won top 10% performer every year with the company.
- Interviewed over a dozen candidates within 3 years.
- Onboarded 3 coworkers in Python best practices and professional software design principles.
- Maintained coding standards, code review, and planning of development milestones.

TECHNICAL SKILLS

Primary Languages: Python, Rust, TypeScript, Bash/Unix, PostgreSQL, Terraform, CloudFormation

Additional Known Languages: Golang, C/C++, C#, Java, Haskell, Clojure, Spark/Scala, Cypher

Technologies: Git, Docker, Kubernetes, Supabase, Redis, MongoDB, DynamoDB, Neo4j, GitHub Actions, AWS, Ansible, Datadog, Linux

COMMUNITY

Open Source Contributions

- [run-llama/llama_index](#)
- [OpenHands/OpenHands](#)
- [guidance-ai/guidance](#)

Open Source Creations (including Archives)

- [ryanpeach/continuing_education](#) (In Progress) - A Python repo where I track ML papers and do replications.
- [ryanpeach/mdlinker](#) (Archived) - Rust executable that runs at git pre-commit to detect and repairs broken wikilinks, duplicate files, and missing aliases in Obsidian and similar markdown note taking apps.
- [ryanpeach/orbitingsandrust](#) (Archived) - A game physics engine I worked on in Rust which I think demonstrates my Rust code quality.

Local Activities

- [PyATL \(Python Atlanta\)](#) – Frequent Speaker on Machine Learning and Data Science in Python.